`knowledge\models\requirements-and-traceability.md`

Requirements and Traceability

Status: draft, owner's review in progress (started 2026-10-10). Written by Claude on 2026-10-10 at the owner's request. Every requirement below has status `draft` in the model. Nothing is baselined, and no box in `projects/nas-sos-capstone/to-do-list.md` section 10 has been checked on the strength of it. Not report text.

#TODO (owner) — review plan, added 2026-10-10 under D-016

Work through these in order. Mark each line done here, then apply the wording to the `.sysml` files (or ask Claude to apply the marked-up copy).

- ☐ **(a) All 27 stakeholder requirements (section 3).** Re-word each in your own language. The test for each: is this what that stakeholder would say they want, and would you defend it to the adviser? Confirm or change the objective rows it traces to. About 3 hours.
- ☐ **(b) The thread's system requirements only (section 4).** These are the ones the experiment and the proposed interface touch: SLR-PLN-02, SLR-PLN-03, SLR-CLR-01, SLR-CLR-02, SLR-CLR-09, SLR-CLR-10, SLR-CLR-12, SLR-CLR-13, SLR-INF-01, SLR-INF-02, SLR-INF-03, SLR-INF-04, SLR-INF-05, SLR-EXE-03, SLR-MEA-02. For each: re-word; give a `measureOfPerformance` and unit where one exists or write "qualitative"; name the verification case that will exercise it (`cameo_models/nas_verification.sysml`, turn 4). Confirm or strike the `[SME]` grades on SLR-CLR-03, SLR-EXE-03 (section 9, item 9).
- ☐ **(c) Write the next tier for the thread** (three to five requirements): what the ATC decision support system and the airline-to-ATC interface shall do, derived from SLR-INF-02/03/04 and SLR-CLR-13, with the constituent system as the subject ("The ATC decision support system shall ..."). This is the worked example of tiering for the report; "The NAS shall" stays the subject of all 53 at the system-of-systems tier (owner's call, 2026-10-10).
- ☐ **(d) Everything else** stays at status `draft` and goes in the report appendix marked unreviewed. Do not spend review time on it before the final report.
- ☐ **(e) Architecture state (D-018).** The five `improved` requirements (SLR-CLR-10, SLR-CLR-13, SLR-INF-02, SLR-INF-03, SLR-INF-04) are now tagged in the model. Confirm that is the complete set.
- ☐ **(f) The three proposed objective rows** (airline data protection, passenger safety, Governance) in `knowledge/models/stakeholder-objective-ontology.md`, so SN-AIR-06, SN-PAX-01 and SN-GOV-01/02 trace (section 9, items 5 and 6).

The model is the source of truth (D-006). This file is a readable copy of two model files, written for review on paper:

- `projects/nas-sos-capstone/cameo_models/requirements_stakeholders.sysml` holds the 27 stakeholder requirements.
- `projects/nas-sos-capstone/cameo_models/requirements_nas_system.sysml` holds the 53 system requirements and their allocation to systems.
- `projects/nas-sos-capstone/cameo_models/requirements_definitions.sysml` holds the attributes both use.

If this file and those disagree, the `.sysml` files win. Mark changes here by hand, then apply them to the `.sysml` files.

1. What is here

- **27 stakeholder requirements** in 8 stakeholder groups. Each says what that group wants from the NAS as a whole, and names the objective rows it comes from in `knowledge/models/stakeholder-objective-ontology.md`.
- **53 system requirements** in 7 areas that follow the lifecycle of trajectory intent (`projects/nas-sos-capstone/index.md`, "Center of gravity").
- **Every stakeholder requirement is answered by at least one system requirement.** Section 3 shows which.
- **48 system requirements trace to a stakeholder requirement. 5 are derived** and carry a rationale. Section 5 lists them.
- **5 system requirements are not met by the system as it operates today** (SLR-CLR-10, SLR-CLR-13, SLR-INF-02, SLR-INF-03, SLR-INF-04). They are the ones the experiment in `knowledge/models/experiment-definition.md` is about.
- Every system requirement is allocated to parts of `NationalAirspaceSystem` in `projects/nas-sos-capstone/cameo_models/nas_sysml_package_definitions.sysml`.

The trace runs: objective row, then stakeholder requirement, then system requirement, then constituent system.

2. How the two example workbooks were used

The owner's earlier workbooks were written for an airline operations center (AOC) as the system. They were used for their form, not their content.

Carried over from `projects/nas-sos-capstone/prework/Stakeholder Requirements.xlsx`, worksheet "Stakeholder Requirements Groupi":

- Requirements are grouped by the stakeholder who wants them.
- Each is one sentence with the system as the subject and one "shall".
- Broad wishes are allowed at this level ("maximize profits", "minimize delays"); the number comes later.
- Rows that repeat another row are merged, as the workbook did with its "already captured" comments.

Carried over from `projects/nas-sos-capstone/prework/1 System Requirements_10-26_v1.xlsx`, worksheet "Report":

- A compound stakeholder requirement is split into single statements (the workbook split "accurate and up-to-date flight plans" into two).
- A vague word is replaced by something that can be checked (the workbook turned "significant changes in fuel status" into "in excess of 5% of expected fuel burned").
- A rule is named by its paragraph where one is known (the workbook named the CFR part for crew fatigue).
- There are more system requirements than stakeholder requirements.

Not carried over:

- The content. Most workbook rows say what the AOC owes another party. Here that other party is usually inside the system, so the row becomes an exchange between two constituent systems.
- Rows about how an airline runs itself (budget, corporate values, IT approval, cybersecurity training, record keeping). They do not touch the trajectory-intent chain.
- Numbers. The workbook's 60 minutes for filing a flight plan and 5 percent fuel deviation are not reused, because no source held in this repository gives them.

How the workbook's stakeholders map onto this project's. Each row states the abstraction and the reason, so it can be defended or changed.

Workbook stakeholder	This project's group	Abstraction	Why
Air Traffic Control (ATC)	Air traffic control and traffic flow management (FAA Air Traffic Organization)	identical	Same stakeholder. The direction reverses: the workbook asked what the AOC owes ATC; here ATC is inside the system, so the rows say what ATC wants from the whole.
Customers	Passengers	renamed	D-007 uses Passengers and treats them as a boundary actor.
Pilots & Flight Crew	Flight crew (captain and first officer)	identical	Cabin crew are left out: no authority over in-scope decisions (nas_context_diagram.sysml, cabinCrew comment).
FAA, EASA & all Aviation Authorities	Regulator (FAA as rule maker; Department of Transportation)	many to one	D-007 scopes to U.S. domestic Part 121, so only the FAA and DOT rules apply.
Airports	Airport operator	identical	
Airline Fleet Planning; Airline Management; Airline & 3rd Party Maintenance & Engineering; Airline Safety & Security Department; Airline IT	Airline (Part 121 scheduled passenger operator)	many to one	In the workbook the AOC was the system, so the rest of the airline stood outside it as separate stakeholders. Here the whole airline is one constituent system (Flight Operations), so its departments are one stakeholder. Their objectives survive as rows in the Airline table of knowledge/models/stakeholder-objective-ontology.md.
Pilot Unions	Flight crew (captain and first officer)	many to one	The union's interest in the workbook was duty and rest limits, which is SN-CRW-05.
(none)	Communities and environmental organizations	gap in the workbook	Added from knowledge/models/stakeholder-objective-ontology.md and knowledge/models/stakeholder-register.md.
(none)	Military airspace users	gap in the workbook	Added from D-007 (boundary actor) and the suaToTfm link in nas_context_diagram.sysml.

Evidence grades used below follow the review key in `projects/nas-sos-capstone/cameo_models/nas_context_diagram.sysml`:

- [A] rule or FAA order held in the source register
- [B] published study held in the source register
- [SME] owner's industry knowledge, to be confirmed by the owner
- [C] general domain knowledge, not yet tied to a source in this repository

Names in `code font` in the Basis column are keys in `evidence/sources/references.bib`. A grade says where the idea comes from. It does not mean the cited paragraph was re-read in the session that wrote this.

3. Stakeholder requirements (in the NAS)

... Provide a means for airlines to ... so that ...

Airline (Part 121 scheduled passenger operator)

D-007 treatment: Modeled system (Flight Operations).

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-AIR-01	The NAS shall let the airline plan and fly each flight on the route, altitude and speed that give that flight its lowest operating cost, within safety and capacity limits.	Airline: Fuel cost; Airline: Delay cost	SLR-PLN-01, SLR-PLN-02, SLR-PLN-03, SLR-CLR-09,	[B] `mori2022massCruise`, `munro2018managingVariability`

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
			SLR-EXE-03, SLR-MEA-02	
SN-AIR-02	The NAS shall bring each flight to its destination gate at its scheduled arrival time.	Airline: Schedule integrity; Airline: Aircraft utilization; Airline: Passenger connections	SLR-INF-07, SLR-APT-01, SLR-APT-04, SLR-MEA-03	[C]
SN-AIR-03	The NAS shall tell the airline about airspace and airport constraints that will delay or reroute its flights early enough for the airline to re-plan.	Airline: Delay cost; Airline: Schedule integrity	SLR-FLW-03, SLR-FLW-07	[A] `faa2025jo72103ee`
SN-AIR-04	The NAS shall let the airline choose which of its own flights takes a delay that the airline has been assigned. *Note: JO 7210.3EE paragraph 18-10-12 (slot substitution) and paragraph 18-4-5 (diversion recovery priorities).*	Airline: Delay cost; Airline: Passenger connections	SLR-FLW-04	[A] `faa2025jo72103ee`
SN-AIR-05	The NAS shall take the airline's priorities for a flight (fuel against time, and protected connections) into account when another party changes that flight's trajectory. *Note: This is the need behind the experiment in knowledge/models/experiment-definition.md section 2. It is not met today: the controller does not receive weight or cost index.*	Airline: Fuel cost; Airline: Schedule integrity; Airline: Passenger connections	SLR-CLR-13, SLR-INF-02, SLR-INF-03, SLR-INF-04	[B] `schultz2012adaptiveClimb`, `coppenbarger1999climbPrediction`
SN-AIR-06	The NAS shall keep the airline's commercially sensitive flight data (aircraft weight, cost index, connection priorities) from other airlines. *Note: No objective row matches this in knowledge/models/stakeholder-objective-ontology.md. Competitive sensitivity is the published reason airlines do not share weight and speed intent. It pulls against SN-AIR-05. Proposed as a new row under Airline.*	none (see note)	SLR-INF-05	[B] `schultz2012adaptiveClimb`

Air traffic control and traffic flow management (FAA Air Traffic Organization)

D-007 treatment: Modeled system (Airspace Management).

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-ATC-01	The NAS shall keep every controlled aircraft separated from every other aircraft by at least the applicable separation minimum.	ATC / ANSP: Safety/separation	SLR-CLR-01, SLR-CLR-02, SLR-CLR-03, SLR-CLR-05	[A] `faa2025jo711065bb`
SN-ATC-02	The NAS shall keep the number of aircraft in each sector and at each airport within its declared capacity.	ATC / ANSP: Capacity; ATC / ANSP: Sector workload; ATC / ANSP: Traffic complexity	SLR-PLN-08, SLR-FLW-01, SLR-FLW-02, SLR-FLW-06, SLR-APT-02	[A] `faa2025jo72103ee`
SN-ATC-03	The NAS shall make the path each controlled aircraft will fly known to its controller in advance.	ATC / ANSP: Predictability	SLR-PLN-06, SLR-FLW-06, SLR-CLR-03, SLR-INF-01, SLR-EXE-01, SLR-EXE-02	[B] `coppenbarger1999climbPrediction`
SN-ATC-04	The NAS shall keep the number of control actions a sector controller must take within what one sector team can sustain.	ATC / ANSP: Sector workload; ATC / ANSP: Traffic complexity	SLR-FLW-02, SLR-CLR-12	[B] `rantanen2012conflictManeuvers`, `kirwan2001coraStrategies`
SN-ATC-05	The NAS shall share unavoidable delay equitably among airspace users. *Note: JO 7210.3EE paragraph 18-10-2: equitable assignment of delays to all system users.*	ATC / ANSP: Delay (ATC-attributed)	SLR-FLW-05, SLR-MEA-04	[A] `faa2025jo72103ee`

Airport operator

D-007 treatment: Modeled system (Airport Operations).

All of these are not D-007

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-APT-01	The NAS shall give the airport operator arrival and departure times reliable enough to assign gates and ground crews.	Airport: Gate utilization; Airport: Turnaround performance	SLR-INF-07, SLR-APT-01	[C]
SN-APT-02	The NAS shall use the runway capacity the airport has declared available.	Airport: Runway utilization	SLR-APT-02	[C]
SN-APT-03	The NAS shall keep the time aircraft wait on taxiways with engines running as short as practical.	Airport: Surface congestion	SLR-APT-03	[C]

Flight crew (captain and first officer)

D-007 treatment: Modeled system (Flight Crew).

1) Provide clearances to flight crews that account for air traffic.
2) Inform crews of weather, traffic, & TMIs
3) Facilitate control of a/c across regional sectors, coordinating controls, clearances, & handoffs.

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-CRW-01	The NAS shall preserve the captain's final authority over the safe conduct of the flight.	Flight Crew: Safety; Flight Crew: Operational flexibility	SLR-PLN-05, SLR-CLR-08, SLR-EXE-04	[A] 14 CFR 91.3
SN-CRW-02	The NAS shall give the flight crew the information needed for each phase of flight before that phase begins.	Flight Crew: Safety; Flight Crew: Procedural compliance	SLR-PLN-01, SLR-PLN-07, SLR-INF-06, SLR-EXE-06	[B] `seamster2011collabSystems`
SN-CRW-03	The NAS shall give the flight crew only trajectory instructions that the aircraft can fly, with enough time to set them up.	Flight Crew: Workload; Flight Crew: Safety	SLR-CLR-10, SLR-CLR-11, SLR-INF-02, SLR-EXE-03	[SME]
SN-CRW-04	The NAS shall let the flight crew ask for a trajectory change and get an answer.	Flight Crew: Operational flexibility	SLR-CLR-09	[B] `engility2014tasarAlaska`
SN-CRW-05	The NAS shall keep each flight crew member within flight and duty time limits.	Flight Crew: Schedule/duty constraints	SLR-PLN-09	[C] 14 CFR Part 117

Not D-007

Passengers

D-007 treatment: Boundary actor.

Not D-007

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-PAX-01	The NAS shall carry passengers without injury. *Note: The Passenger table in knowledge/models/stakeholder-objective-ontology.md has no safety row. The interest comes from knowledge/models/stakeholder-register.md (Social: Passengers, safety and convenience). Proposed as a new Passenger row.*	Airline: Flight Safety	SLR-PLN-04, SLR-CLR-01, SLR-CLR-08, SLR-EXE-06	[C]
SN-PAX-02	The NAS shall get passengers to their destination, including connections, at the time they were sold.	Passenger: Travel time; Passenger: Connection reliability; Passenger: Disruption risk	SLR-INF-04, SLR-MEA-03	[C]
SN-PAX-03	The NAS shall avoid known turbulence where a route or altitude change allows it.	Passenger: Comfort	SLR-CLR-09, SLR-INF-06	[C]

Regulator (FAA as rule maker; Department of Transportation)

D-007 treatment: Boundary actor.

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-GOV-01	The NAS shall conduct every flight within the operating rules of 14 CFR and the FAA's air traffic orders. *Note: Governance has no table in knowledge/models/stakeholder-objective-ontology.md. The objective is 'Regulatory Compliance' in the enterprise objective hierarchy in knowledge/models/stakeholder-register.md.*	none (see note)	SLR-PLN-04, SLR-PLN-05, SLR-PLN-09	[A] `faa2025jo711065bb`, `faa2025jo72103ee`, 14 CFR 91, 14 CFR 121
SN-GOV-02	The NAS shall make one identified authority accountable for each decision that changes a flight's trajectory. *Note: From 'Regulatory Compliance' and 'Safety' in the enterprise objective hierarchy in knowledge/models/stakeholder-register.md. The joint dispatcher*	none (see note)	SLR-PLN-05, SLR-CLR-05, SLR-EXE-04, SLR-MEA-01	[A] 14 CFR 91.3, 14 CFR 121.533, `faa2025jo711065bb`

IDK about this one... the Note is an exception!

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
	and captain release is the one place the rules deliberately share authority.			

Communities and environmental organizations

D-007 treatment: Boundary actor (acts through Governance per D-007).

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-ENV-01	The NAS shall keep the CO2 emitted by each flight as low as safety and capacity allow.	Environmental / Societal: CO2	SLR-PLN-03, SLR-CLR-13, SLR-APT-03, SLR-MEA-02	[C] `icao2024icecMethodology`
SN-ENV-02	The NAS shall keep aircraft noise over communities near airports within the published noise abatement procedures.	Environmental / Societal: Noise; Environmental / Societal: Community impacts	SLR-APT-05	[C]

Military airspace users

D-007 treatment: Boundary actor.

ID	The stakeholder wants	Objective rows it comes from	Answered by	Basis
SN-MIL-01	The NAS shall give military operations sole use of special use airspace while it is scheduled active.	Military: Airspace access	SLR-PLN-08, SLR-FLW-07, SLR-CLR-14	[C]

4. System requirements

Columns: **Traces to** is the stakeholder requirement it answers, or DERIVED. **Allocated to** names the constituent systems, the accountable one first. **Check** is how it would be verified: inspection of the model, analysis (calculation or simulation), or demonstration (a scenario walked through on the activity diagram). **Measure** is filled only where there is something to measure; a number appears only where a source gives it.

Flight planning and release

Place on the trajectory-intent chain: mission plan -> flight plan.

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
SLR-PLN-01	The NAS shall produce for each scheduled flight a flight plan that states its route, cruise altitude, cruise speed and fuel load.	Functional	SN-AIR-01, SN-CRW-02	Dispatcher	Inspection		[B] `berry2011aocActors`
SLR-PLN-02	The NAS shall compute each flight plan from the planned take-off weight and the performance of the aircraft assigned to the flight.	Functional	SN-AIR-01	AOC decision support (flight planning system); Load planner	Analysis		[B] `mori2022massCruise`
SLR-PLN-03	The NAS shall select each flight plan's route, altitude and speed to minimize the airline's stated cost of fuel and flight time.	Performance	SN-AIR-01, SN-ENV-01	AOC decision support (flight planning system)	Analysis	flight cost = fuel cost + cost index x flight time (USD per flight)	[B] `mori2022massCruise`
SLR-PLN-04	The NAS shall load each flight with fuel to reach its destination, then its most distant alternate, then fly 45 minutes at normal cruise.	Constraint	SN-GOV-01, SN-PAX-01	Dispatcher	Inspection	fuel remaining after destination and alternate (minutes at normal cruise). Value: 45	[C] 14 CFR 121.639
SLR-PLN-05	The NAS shall release a flight only when the dispatcher and the captain both agree that it	Functional	SN-CRW-01, SN-GOV-01,	Dispatcher; Flight deck crew	Inspection		[A] 14 CFR 121.533

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
	can be flown safely as planned.		SN-GOV-02				
SLR-PLN-06	The NAS shall file each flight plan with air traffic control before the flight departs.	Interface	SN-ATC-03	Dispatcher; Air traffic control	Inspection		[A] `faa2025aim`, `faa2025jo72103ee`
SLR-PLN-07	The NAS shall deliver the dispatch release, flight plan, weather, notices to airmen (NOTAMs) and load sheet to the flight crew before departure.	Interface	SN-CRW-02	Dispatcher; Load planner	Inspection		[B] `seamster2011collabSystems`
SLR-PLN-08	The NAS shall plan each flight to comply with the traffic management initiatives and special use airspace activations published for its time of flight.	Constraint	SN-ATC-02, SN-MIL-01	Dispatcher; AOC ATC coordinator	Inspection		[A] `faa2025jo72103ee`
SLR-PLN-09	The NAS shall assign to each flight a crew whose duty period, including forecast delay, stays within the limits of 14 CFR Part 117.	Constraint	SN-CRW-05, SN-GOV-01	Crew scheduler	Inspection		[C] 14 CFR Part 117

Not important to project.

Traffic flow management

TMIs aren't what I'm studying. Should they be?

Place on the trajectory-intent chain: flight plan -> ATC constraints (before departure and strategic).

Miles/Minutes in trail

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
SLR-FLW-01	The NAS shall forecast traffic demand against declared capacity for each airport and each constrained volume of airspace.	Functional	SN-ATC-02	Command Center (ATCSCC); Traffic management unit	Inspection		[A] `faa2025jo72103ee`
SLR-FLW-02	The NAS shall issue a traffic management initiative when forecast demand exceeds declared capacity.	Functional	SN-ATC-02, SN-ATC-04	Command Center (ATCSCC)	Demonstration		[A] `faa2025jo72103ee`
SLR-FLW-03	The NAS shall notify each affected airline of a traffic management initiative before its affected flights depart.	Interface	SN-AIR-03	Command Center (ATCSCC); AOC ATC coordinator	Inspection		[A] `faa2025jo72103ee`, `berry2011aocActors`
SLR-FLW-04	The NAS shall let an airline exchange its own flights among the departure slots it has been assigned.	Functional	SN-AIR-04	Command Center (ATCSCC); AOC ATC coordinator	Demonstration		[A] `faa2025jo72103ee`
SLR-FLW-05	The NAS shall assign the delay created by a traffic management initiative equitably among all airspace users.	Performance	SN-ATC-05	Command Center (ATCSCC)	Analysis	assigned delay per flight, compared across airlines (minutes)	[A] `faa2025jo72103ee`
SLR-FLW-06	The NAS shall release each flight that has an expect departure clearance time (EDCT) within the permitted window around that time.	Performance	SN-ATC-02, SN-ATC-03	Control tower; Flight deck crew	Analysis	actual departure time minus EDCT (minutes)	[C] `faa2025jo72103ee`

Sub-optimal... I don't think I want to.

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
SLR-FLW-07	The NAS shall give traffic flow managers and airlines the activation schedule of special use airspace before the affected flight plans are filed.	Interface	SN-MIL-01, SN-AIR-03	Command Center (ATCSCC)	Inspection		[C]

Clearance, separation and tactical trajectory change

Place on the trajectory-intent chain: ATC constraints -> trajectory negotiation.

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
SLR-CLR-01	The NAS shall keep each pair of controlled aircraft separated by at least the minimum that JO 7110.65BB sets for the airspace they are in.	Performance	SN-ATC-01, SN-PAX-01	Air traffic control; En-route center (ARTCC); Terminal approach control (TRACON)	Analysis	lateral and vertical distance between each pair of aircraft (NM and ft)	[A] `faa2025jo711065bb`
SLR-CLR-02	The NAS shall predict a loss of separation between controlled aircraft before it occurs.	Functional	SN-ATC-01	ATC decision support	Analysis		[B] `engility2014tasarAlaska`
SLR-CLR-03	The NAS shall give the controlling facility the position, altitude and identity of each controlled aircraft.	Interface	SN-ATC-01, SN-ATC-03	Aircraft surveillance system; Air traffic control	Inspection		[SME]
SLR-CLR-04	The NAS shall provide a two-way communication path between the controlling facility and the flight crew for the whole of controlled flight.	Interface	**DERIVED** from SLR-CLR-05, SLR-CLR-08, SLR-CLR-09	Aircraft communication system; Air traffic control	Inspection		[SME]
SLR-CLR-05	The NAS shall change a controlled aircraft's cleared route, altitude or speed only by a clearance from the facility that has control of that aircraft.	Functional	SN-GOV-02, SN-ATC-01	Air traffic control	Inspection		[A] `faa2025jo711065bb`
SLR-CLR-06	The NAS shall have the flight crew read back or acknowledge each clearance	Functional	**DERIVED** from SLR-CLR-01, SLR-CLR-05	Flight deck crew; Air traffic control	Demonstration		[C] `seamster2011collabSystems`

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
	before acting on it.						
SLR-CLR-07	The NAS shall transfer control of an aircraft from one sector or facility to the next only after the receiving controller has accepted it.	Functional	**DERIVED** from SLR-CLR-01, SLR-CLR-05	En-route center (ARTCC); Terminal approach control (TRACON); Control tower	Demonstration		[A] `faa2025jo711065bb`
SLR-CLR-08	The NAS shall allow the captain to deviate from a clearance when safety requires it, with air traffic control told as soon as practical.	Functional	SN-CRW-01, SN-PAX-01	Flight deck crew	Inspection		[A] 14 CFR 91.3
SLR-CLR-09	The NAS shall answer each flight crew request for a route, altitude or speed change with an approval, an amended clearance or a refusal.	Functional	SN-CRW-04, SN-AIR-01, SN-PAX-03	Air traffic control; Flight deck crew	Demonstration		[B] `engility2014tasarAlaska`
SLR-CLR-10	The NAS shall issue only clearances that the aircraft can fly at its current weight. **Not met today.**	Constraint	SN-CRW-03	Air traffic control	Analysis		[B] `coppenbarger1999climbPrediction`, `engility2014tasarAlaska`
SLR-CLR-11	The NAS shall issue holding instructions at least 5 minutes before the aircraft reaches its clearance limit when a delay is expected.	Performance	SN-CRW-03	Air traffic control	Inspection	time between the holding instruction and arrival at the clearance limit (minutes). Value: 5	[A] `faa2025jo711065bb`, `seamster2011collabSystems`
SLR-CLR-12	The NAS shall resolve each predicted conflict with the fewest control instructions that restore separation.	Performance	SN-ATC-04	Air traffic control	Analysis	instructions issued plus follow-up instructions, per conflict (count)	[B] `rantanen2012conflictManeuvers`, `kirwan2001coraStrategies`
SLR-CLR-13	The NAS shall choose, among the maneuvers that restore separation with the same number of control instructions, the one with	Performance	SN-AIR-05, SN-ENV-01	ATC decision support; Air traffic control	Analysis	sum over the aircraft involved of fuel cost plus cost index x time (USD per conflict)	[B] `schultz2012adaptiveClimb`, `mori2022massCruise`

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
	the lowest combined cost to the aircraft involved. **Not met today.**						
SLR-CLR-14	The NAS shall keep civil aircraft out of special use airspace while it is active, unless the using agency releases it.	Functional	SN-MIL-01	Air traffic control	Inspection		[C] faa2025jo711065bb

Information for trajectory decisions

Place on the trajectory-intent chain: what the party making each decision must be given.

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
SLR-INF-01	The NAS shall give the sector controller the filed route, requested altitude and requested speed of each aircraft before it enters the sector.	Interface	SN-ATC-03	Air traffic control; En-route center (ARTCC)	Inspection		[A] faa2025jo711065bb
SLR-INF-02	The NAS shall give the party that decides a change to an aircraft's cruise altitude that aircraft's current gross weight. **Not met today.**	Interface	SN-AIR-05, SN-CRW-03	Flight management system; Airline operations center; Air traffic control	Analysis		[B] schultz2012adaptiveClimb, coppenbarger1999climbPrediction
SLR-INF-03	The NAS shall give the party that decides a change to an aircraft's cruise altitude or speed that flight's cost index. **Not met today.**	Interface	SN-AIR-05	Flight management system; Airline operations center; Air traffic control	Analysis		[B] schultz2012adaptiveClimb
SLR-INF-04	The NAS shall give the party that decides a delay or trajectory change for a flight the airline's connection priority for that flight. **Not met today.**	Interface	SN-AIR-05, SN-PAX-02	Airline operations center; Command Center (ATCSCC); Air traffic control	Analysis		[SME]
SLR-INF-05	The NAS shall disclose an airline's shared weight, cost index and connection priority only to the parties deciding that flight's trajectory.	Constraint	SN-AIR-06	Air traffic control; Command Center (ATCSCC)	Inspection		[B] schultz2012adaptiveClimb
SLR-INF-06	The NAS shall give the dispatcher and the flight crew the reported and forecast turbulence along the planned route.	Interface	SN-PAX-03, SN-CRW-02	Airline operations center; Flight deck crew	Inspection		[C] seamster2011collabSystems

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
SLR-INF-07	The NAS shall send the airline and the destination airport an updated arrival estimate whenever a flight's estimate changes by more than 5 minutes.	Interface	SN-AIR-02, SN-APT-01	Flight management system; Flight deck crew; Airline operations center; Airport management system	Inspection	change in estimated arrival time that triggers an update (minutes). Value: 5	[B] `seamster2011collabSystems`
SLR-INF-08	The NAS shall label every exchanged flight plan, clearance, surveillance report and arrival estimate with one flight identifier and one time reference shared by all parties.	Interface	**DERIVED** from SLR-PLN-06, SLR-CLR-03, SLR-INF-01, SLR-INF-07	Airline operations center; Air traffic control; Aircraft surveillance system	Inspection		[C] `faaFixmUsExtension2024`

Sure, but why relevant for my project?

Onboard execution

Place on the trajectory-intent chain: FMS intent -> guidance commands -> aircraft motion.

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
SLR-EXE-01	The NAS shall hold each flight's cleared route and altitudes in the flight management system as the path the aircraft's guidance follows.	Functional	SN-ATC-03	Flight management system	Inspection		[SME]
SLR-EXE-02	The NAS shall fly each aircraft along its cleared route and at its cleared altitude within the navigation and altitude tolerances of the airspace.	Performance	SN-ATC-03	Auto flight control system; Flight management system	Analysis	cross-track distance from the cleared route; difference from the cleared altitude (NM and ft)	[SME]
SLR-EXE-03	The NAS shall predict on board each flight's arrival time, fuel at destination, optimum altitude and maximum altitude from the aircraft's actual weight, the forecast wind and the cost index.	Functional	SN-AIR-01, SN-CRW-03	Flight management system	Inspection		[SME] `mori2022massCruise`
SLR-EXE-04	The NAS shall change an aircraft's flown trajectory only through flight deck controls operated by the flight crew.	Constraint	SN-CRW-01, SN-GOV-02	Flight deck controls; Flight deck crew	Inspection		[SME] 14 CFR 91.3
SLR-EXE-05	The NAS shall determine each aircraft's position on board with the accuracy its cleared route requires.	Performance	**DERIVED** from SLR-EXE-02, SLR-CLR-03	Navigation system	Inspection		[SME]
SLR-EXE-06	The NAS shall alert the flight crew when the fuel predicted at the destination falls below the required reserve.	Functional	SN-PAX-01, SN-CRW-02	Flight management system	Inspection		[SME]

Airport and surface

Place on the trajectory-intent chain: start and end of the chain on the ground.

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
SLR-APT-01	The NAS shall assign each arriving flight a gate before it lands.	Functional	SN-APT-01, SN-AIR-02	Airport management system	Demonstration		[C]
SLR-APT-02	The NAS shall sequence arrivals and departures so that each runway is used at its declared rate whenever demand is at or above that rate.	Performance	SN-APT-02, SN-ATC-02	Terminal approach control (TRACON); Control tower	Analysis	runway operations per hour against the declared rate (operations per hour)	[A] faa2025jo72103ee
SLR-APT-03	The NAS shall hold a flight that has a known departure delay at its gate with engines off, not in a taxiway queue.	Functional	SN-APT-03, SN-ENV-01	Control tower; Airline operations center	Demonstration		[C]
SLR-APT-04	The NAS shall have each aircraft fuelled, loaded and closed up by its scheduled off-block time.	Performance	SN-AIR-02	Ground handling; Fuel distribution system; Flight deck crew	Demonstration	actual off-block time minus scheduled off-block time (minutes)	[B] schultz2017turnaround
SLR-APT-05	The NAS shall assign departure and arrival procedures that follow the airport's published noise abatement procedures.	Constraint	SN-ENV-02	Control tower; Terminal approach control (TRACON); Dispatcher	Inspection		[B] seamster2011collabSystems

Measurement and records

Place on the trajectory-intent chain: back up the chain: what is reported to the authorities and stakeholders.

ID	Requirement	Kind	Traces to	Allocated to	Check	Measure	Basis
SLR-MEA-01	The NAS shall record each dispatch release, clearance and traffic management initiative with the authority that issued it and the time.	Functional	SN-GOV-02	Dispatcher; Air traffic control; Command Center (ATCSCC)	Inspection		[A] faa2025jo72103ee
SLR-MEA-02	The NAS shall report the fuel burned and the CO2 emitted by each flight, at 3.16 kg of CO2 per kg of fuel.	Performance	SN-ENV-01, SN-AIR-01	Flight management system; Airline operations center	Analysis	CO2 emitted per kg of fuel burned (kg CO2 per kg fuel). Value: 3.16	[A] icao2024icecMethodology
SLR-MEA-03	The NAS shall report each flight's gate arrival time against its scheduled arrival time.	Functional	SN-AIR-02, SN-PAX-02	Airline operations center	Analysis	gate arrival time minus scheduled arrival time (minutes)	[C]
SLR-MEA-04	The NAS shall report the delay each flight received from traffic management initiatives, by airline.	Functional	SN-ATC-05	Command Center (ATCSCC)	Analysis	assigned delay per flight (minutes)	[C]

5. Derived system requirements and why each exists

A derived requirement has no stakeholder behind it. It is there because of a design choice, or because other system requirements cannot be met without it. Each one names those other requirements.

SLR-CLR-04. The NAS shall provide a two-way communication path between the controlling facility and the flight crew for the whole of controlled flight.

- Needed by: SLR-CLR-05, SLR-CLR-08, SLR-CLR-09.

- Why it exists: No stakeholder asks for a radio or a datalink. It exists because a clearance (SLR-CLR-05), a notice of an emergency deviation (SLR-CLR-08) and a crew request (SLR-CLR-09) all have to cross between the ground and the flight deck. Without it none of the three can be met.

SLR-CLR-06. The NAS shall have the flight crew read back or acknowledge each clearance before acting on it.

- Needed by: SLR-CLR-01, SLR-CLR-05.
- Why it exists: A clearance that is misheard is flown as a different trajectory from the one the controller separated. The read-back is the check that the clearance issued and the clearance received are the same. Without it SLR-CLR-01 rests on an unchecked message. No stakeholder states this as a need; it follows from choosing voice and datalink messages as the way SLR-CLR-05 is carried out.

SLR-CLR-07. The NAS shall transfer control of an aircraft from one sector or facility to the next only after the receiving controller has accepted it.

- Needed by: SLR-CLR-01, SLR-CLR-05.
- Why it exists: A controller's authority ends at the sector boundary (knowledge/models/stakeholder-personas.md, Air Traffic Controller, Decision authority). A flight crosses many sectors, so without an accepted handoff there would be a stretch where no facility is accountable for its separation, which breaks SLR-CLR-05. It follows from dividing the airspace into sectors, which is a design choice and not a stakeholder need.

SLR-INF-08. The NAS shall label every exchanged flight plan, clearance, surveillance report and arrival estimate with one flight identifier and one time reference shared by all parties.

- Needed by: SLR-PLN-06, SLR-CLR-03, SLR-INF-01, SLR-INF-07.
- Why it exists: The plan, the clearance, the surveillance track and the arrival estimate are produced by different systems. Unless they name the flight and the time in the same way, the receiving party cannot tell that they describe the same aircraft. No stakeholder asks for this; it is a property the exchanges need once four systems are involved. faaFixmUsExtension2024 is a candidate source and has not been checked for this point.

SLR-EXE-05. The NAS shall determine each aircraft's position on board with the accuracy its cleared route requires.

- Needed by: SLR-EXE-02, SLR-CLR-03.
- Why it exists: An aircraft cannot stay on a cleared path (SLR-EXE-02) or report where it is (SLR-CLR-03) unless it knows its own position. No stakeholder asks for navigation as such; both of those requirements fail without it. The ground navigation aids it may use are Infrastructure, a context constraint under D-007, so only the onboard function is required here.

6. Notes on individual system requirements

Requirements that trace to a stakeholder but carry a caution: a number not yet confirmed, a threshold left open, or a statement that today's system does not meet.

- **SLR-PLN-02**: The cost-optimal altitude depends on mass (mori2022massCruise, Figure 1), so a plan built on a nominal weight is not the lowest-cost plan for the aircraft that flies it.
- **SLR-PLN-04**: Number and paragraph are from general knowledge of the domestic fuel supply rule. Confirm against 14 CFR 121.639 before baselining.
- **SLR-PLN-06**: No lead time is set. The summary of faa2025aim records a 45 to 46 minute point before departure after which a filed plan is locked; whether that is the right threshold here is the owner's call.
- **SLR-FLW-04**: JO 7210.3EE paragraph 18-10-12.
- **SLR-FLW-05**: JO 7210.3EE paragraph 18-10-2. What counts as equitable is not quantified in this draft.
- **SLR-FLW-06**: The window is commonly given as 5 minutes before to 5 minutes after the EDCT. That figure is from general knowledge and is left out of the threshold until it is confirmed against JO 7210.3EE or JO 7110.65BB.
- **SLR-CLR-01**: A hard constraint, not something to trade (knowledge/models/experiment-definition.md section 6). For en-route radar the minimum is commonly 5 NM or 1,000 ft; those numbers are from general knowledge and are left out of the threshold until the paragraph is confirmed.
- **SLR-CLR-02**: No look-ahead time is set. engility2014tasarAlaska page 9 uses an eight-minute probe in its own model, which is a candidate.
- **SLR-CLR-10**: Not fully met today. The ground system assumes one nominal weight per aircraft type (coppenbarger1999climbPrediction), and the crew answers 'unable' when a clearance cannot be flown. The rule in D-013 (offer a climb only at or below FL350) stands in for the missing weight. Depends on SLR-INF-02.
- **SLR-CLR-11**: JO 7110.65BB paragraph 4-6-1, as recorded in evidence/literature-notes/annotations/seamster2011collabSystems-bb-rules.csv.
- **SLR-CLR-13**: Not met today, because the party choosing the maneuver lacks the inputs in SLR-INF-02 and SLR-INF-03. The experiment in knowledge/models/experiment-definition.md measures what meeting it is worth. Whether this belongs in the baseline or is a proposed improvement is the owner's call.
- **SLR-INF-02**: Not met today: the controller does not receive weight (schultz2012adaptiveClimb, page 2). This is case C of the experiment (knowledge/models/experiment-definition.md section 4). Which party should hold the decision, and so receive the weight, is still open in knowledge/questions/open-questions.md.
- **SLR-INF-03**: Not met today (schultz2012adaptiveClimb, page 2). Case C of the experiment.
- **SLR-INF-04**: Case D of the experiment (D-014 item 4). Cost index understates a connection-critical flight. The supporting dispatcher-ratings paper (Sheth et al.) is held in evidence/sources/ but is not yet registered, so no key is cited.
- **SLR-INF-07**: The 5 minutes is the crew-to-dispatch practice in seamster2011collabSystems (survey item: notify dispatch of ETA changes greater than 5 minutes). Extending the same trigger to the airport is this draft's proposal, not the source's.
- **SLR-EXE-02**: No tolerance is set. It depends on the navigation specification of the route, which the scenario has not chosen.

- **SLR-EXE-03**: This is the item FmsPerformancePrediction in nas_sysml_package_definitions.sysml. It is the source of what SLR-INF-02 would share.
- **SLR-EXE-04**: D-012: the crew acts through the flight deck controls. No ground party commands the aircraft directly, which is what keeps the captain's authority real.
- **SLR-MEA-02**: icao2024icecMethodology page 6, accepted in D-015. The number is a conversion factor, not a limit.
- **SLR-MEA-03**: A flight is commonly counted on time within 15 minutes of schedule. That figure is from general knowledge and is left out of the threshold until it is sourced.
- **SLR-MEA-04**: SLR-FLW-05 cannot be checked without it.

7. Objectives with no requirement

Rows of `knowledge/models/stakeholder-objective-ontology.md` that no stakeholder requirement picks up, and the reason.

Category	Objective rows	Why not carried
Airline	Crew cost; Maintenance; Revenue; Dispatch reliability	The ontology's own 'Trajectory impact' column marks these Indirect, and its abstraction comment keeps them at schedule level or as boundary inputs. Crew duty limits are picked up through SN-CRW-05.
Passenger	Ticket price; Schedule convenience	Ticket price is marked 'No' trajectory impact; schedule convenience is set when the timetable is designed.
Airport	Passenger throughput	Marked 'No' trajectory impact: a terminal process.
Environmental / Societal	NOx; Contrail/climate effects; Local air quality	The ontology calls this table the least grounded, and no source is held. D-015 notes the altitude link to non-CO2 effects still needs a source. Left out, not judged unimportant.
Military	Mission effectiveness; Mission timing; Fuel availability; Operational security; Resilience	D-007 makes military operations a boundary actor. Only 'Airspace access' crosses into the civil system.

8. What became of the earlier placeholders

The two requirement files held ten needs and six system requirements written on 2026-09-19 as placeholders. They are replaced. The old text is in git history.

Earlier placeholder	Now	Reason
SN-001 regulatory compliance	SN-GOV-01	Kept, narrowed to operating rules and air traffic orders.
SN-002 support new regulations	not carried	D-007 holds regulation fixed in the baseline.
SN-003 efficient and reliable air traffic management	SN-AIR-01 to SN-AIR-05	Split: one sentence held five separate things an airline wants.
SN-004 controller situational awareness and decision support tools	SN-ATC-03, SN-ATC-04	Restated as the need. Decision support tools are a solution, owned per actor under D-012.
SN-005 coordination between ATC centers	SLR-CLR-07 (derived)	Coordination between centers follows from dividing the airspace into sectors; it is a derived system requirement, not a stakeholder need.
SN-006 platform for new aircraft and engine technology	not carried	Manufacturers are not in the seven categories of knowledge/models/stakeholder-objective-ontology.md and have no row in D-007. The group stays in the schema, empty, in case the owner wants it back.
SN-007 airport operations support	SN-APT-01	Kept, restated as reliable times.
SN-008 flight crew decision support	SN-CRW-02, SN-CRW-03	Restated as the need.
SN-009 passenger safety and comfort	SN-PAX-01, SN-PAX-03	Split into safety and ride comfort.
SN-010 environmental data and insight	SN-ENV-01, SN-ENV-02, SLR-MEA-02	The need is low CO2 and noise; reporting data is the system requirement.
SLR-001 real-time monitoring and control	SLR-CLR-01, SLR-CLR-03	
SLR-002 controller-pilot communication	SLR-CLR-04 (derived)	
SLR-003 navigation aids	SLR-EXE-05 (derived)	Ground navigation aids are Infrastructure, a context constraint under D-007.
SLR-004 surveillance	SLR-CLR-03	
SLR-005 coordination between ATC centers	SLR-CLR-07 (derived)	
SLR-006 conflict detection and resolution	SLR-CLR-02, SLR-CLR-12,	

Earlier placeholder	Now	Reason
	SLR-CLR-13	

9. Open items for the owner

1. **Read every requirement and change the wording.** The statements are Claude's. The judgment about what each stakeholder actually wants is the owner's, and so is the defence of it.
2. **Subject of the system requirements.** Each one says "The NAS shall", with the constituent systems named in the allocation. The alternative is to make the constituent system the subject ("Air traffic control shall"). That reads more directly but turns these into requirements on the domains, which `projects/nas-sos-capstone/cameo_models/sysmlv2_exploration.md` section 4 treats as the next level down. Turn them into HLRs
3. **Are the 5 requirements that are not met today requirements, or proposals?** SLR-CLR-10, SLR-CLR-13, SLR-INF-02, SLR-INF-03, SLR-INF-04 describe an improved system, not the current one. If the requirements model is meant to describe the current NAS, they belong in a separate "proposed" set. This ties to the central claim still open in `knowledge/models/experiment-definition.md` section 1.
4. **Two stakeholder requirements pull against each other on purpose.** SN-AIR-05 asks that the airline's priorities be used; SN-AIR-06 asks that the data carrying them be kept from competitors. SLR-INF-05 is the proposed reconciliation. Impact of my changes = ?
5. **Two objective rows are proposed for `knowledge/models/stakeholder-objective-ontology.md`:** protection of shared airline data (Airline), and safety (Passenger). SN-AIR-06 has no matching row at all. SN-PAX-01 borrows the Airline row "Flight Safety" because the Passenger table has no safety row. Both stay that way until the rows are added or the requirements are dropped. The ontology file was not edited. Not relevant to D-007.
6. **Governance has no objective table** in `knowledge/models/stakeholder-objective-ontology.md`. SN-GOV-01 and SN-GOV-02 trace to the enterprise objective hierarchy in `knowledge/models/stakeholder-register.md` instead. Not important.
7. **Numbers to confirm before baselining.** 45 minutes of reserve fuel (SLR-PLN-04, 14 CFR 121.639) is set as a threshold from general knowledge. Left out of the thresholds on purpose and mentioned only in the notes: the EDCT window (SLR-FLW-06), en-route separation of 5 NM or 1,000 ft (SLR-CLR-01), on time within 15 minutes (SLR-MEA-03), a filing lead time (SLR-PLN-06), a conflict look-ahead time (SLR-CLR-02), path conformance tolerances (SLR-EXE-02).
8. **Requirements graded [C]** rest on general knowledge and need a source or the owner's [SME] statement. System: SLR-PLN-04, SLR-PLN-09, SLR-FLW-06, SLR-FLW-07, SLR-CLR-06, SLR-CLR-14, SLR-INF-06, SLR-INF-08, SLR-APT-01, SLR-APT-03, SLR-MEA-03, SLR-MEA-04. Stakeholder, graded [C] or [SME]: SN-AIR-02, SN-APT-01, SN-APT-02, SN-APT-03, SN-CRW-03, SN-CRW-05, SN-PAX-01, SN-PAX-02, SN-PAX-03, SN-ENV-01, SN-ENV-02, SN-MIL-01.
9. **Requirements graded [SME]** were written on the assumption that the owner can vouch for them from industry experience. Confirm or strike: SLR-CLR-03, SLR-CLR-04, SLR-INF-04, SLR-EXE-01, SLR-EXE-02, SLR-EXE-03, SLR-EXE-04, SLR-EXE-05, SLR-EXE-06.
10. **Allocation is a first proposal.** In particular SLR-INF-02 to SLR-INF-04 are allocated to the sender (flight management system or AOC) and to air traffic control as the receiver, which assumes the controller stays the party that decides. `knowledge/questions/open-questions.md` still asks which block should own that.
11. **Manufacturers** have no requirements. The earlier placeholder SN-006 was not carried over.
12. **Cabin crew, general aviation, cargo and international operations** have no requirements, following D-007.

10. Coverage check

Run when these files were generated on 2026-10-10:

- Stakeholder requirements with no system requirement: 0 of 27.
- System requirements with neither a stakeholder parent nor a rationale: 0 of 53.
- System requirements with no allocation: 0 of 53.
- Objective rows named that do not exist in `knowledge/models/stakeholder-objective-ontology.md`: 0.
- Source keys named that do not exist in `evidence/sources/references.bib`: 0.
- `syside check` over the whole `projects/nas-sos-capstone/cameo_models/` folder: no errors.